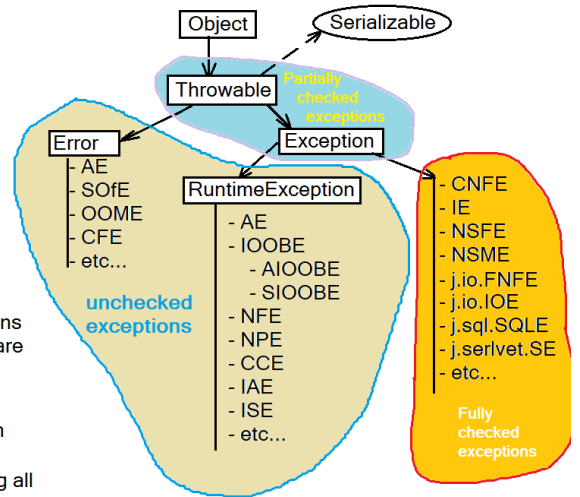


Summary on types of exceptions

- 1) Unchecked exceptions
- 2) Checked exceptions
 - Partially checked
 - Fully checked

- Error and its sub classes, RuntimeException and its sub classes are called unchecked exceptions. Because when we throw those exceptions, compiler will not check for those exceptions handling and doesn't throw error.
- Except Error, RuntimeException and their sub classes, all other exception classes means Throwable, Exception and their direct sub classes are checked exceptions. because when we throw those exceptions, compiler checks their handling. If we don't handle either by reporting or by catching, compiler will throw error. Moreover those exceptions are thrown by programmer explicitly by checking one condition. Because those exceptions are thrown by checking and compiler checks its handling, those exceptions are named as checked exceptions
- checked exceptions are divided into two types partially checked and fully checked based on their sub classes type and based on their rules.
 - Throwable and Exception classes are partially checked exception classes and remaining all direct sub classes of Throwable and Exception are fully checked exception classes.
 - The exception classes those have both unchecked and checked exception classes as sub classes and if some of the rules of checked exception are not applied, those classes are called as partially checked exception classes. Throwable and Exception classes have unchecked exceptions Error and RuntimeException classes as sub classes and also one checked exception rule is not applied that is we can catch Throwable and Exception objects without throwing them from corresponding try statement hence they are called partially checked exceptions
 - The exception classes those do not have unchecked exception class as sub class and if all checked exception rules are applied, we call those exception classes as fully checked exception classes. Throwable and Exception direct sub classes, except Error and RuntimeException classes, are called fully checked exception classes. For example ClassNotFoundException, InterruptedException, IOException, SQLException, ServletException, etc... are fully checked exceptions



Q) Can we call checked exceptions as compile time exceptions

A) Noooooo, at compile time we will get only errors, syntax errors. We will get exceptions only at runtime, because either wrong inputs or because of wrong logic.

Checked exception are checked by compiler, if they are not handle compiler will throw error, it is syntax mistake. Not handle checked exception is also consider as syntax mistake and compiler throws error; unreported exception must be caught or declared to be thrown